

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA4301

COURSE NAME: Internet Programming

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Assessment Tool : Case Study

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Question 1: Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. `<video src="video.mp4" controls></video>`

Explanation

The `<video>` element is the standard HTML5 tag used to embed video content directly into a webpage. It supports various video formats and provides built-in playback controls when the controls attribute is included.

Sample Code

```
<!DOCTYPE html>
<html>
<head>
  <title>Video Example</title>
</head>
<body>

<h2>Promotional Video</h2>

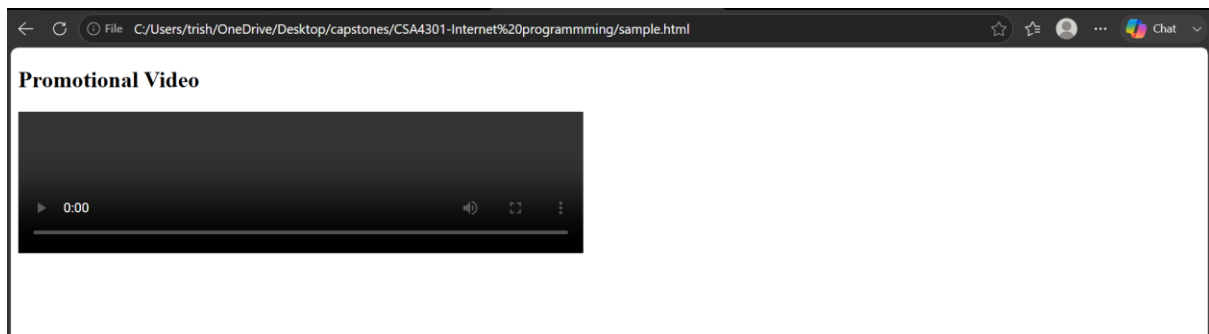
<video src="video.mp4" controls width="600">
  Your browser does not support HTML5 video.
</video>

</body>
</html>
```

Why the Other Options Are Incorrect

Option	Reason
<media>	Not a valid HTML element.
<movie>	Does not exist in HTML5.
<vid>	Invalid HTML tag.

Output Screen:



Question 2: Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Explanation

The POST method is used when submitting sensitive information such as:

- User registration
- Login credentials
- Passwords
- Personal details

POST sends the data inside the HTTP request body instead of the URL, making it more secure than GET.

Sample HTML Form

```
<form action="register.php" method="POST">
```

```
<label>Name:</label>
```

```
<input type="text" name="username">
```

```
<label>Password:</label>
```

```
<input type="password" name="password">
```

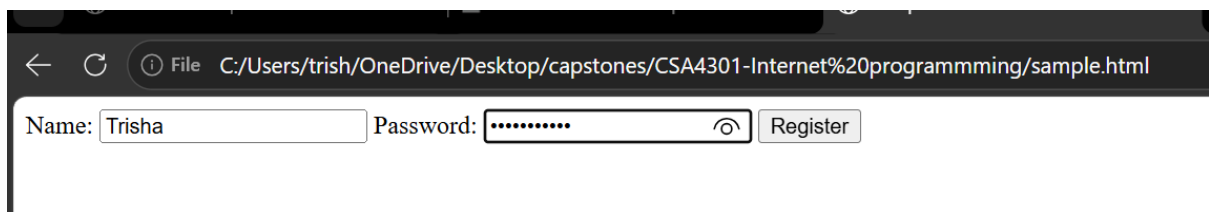
```
<input type="submit" value="Register">
```

```
</form>
```

Comparison of HTTP Methods

Method	Purpose	Suitable for Registration
GET	Retrieve data	✗ No
POST	Send secure data	✓ Yes
PUT	Update existing resource	✗ No
DELETE	Delete resource	✗ No

Sample Output:



Question 3: Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

Explanation

The <nav> element defines a section containing navigation links.

Using semantic HTML improves:

- Accessibility
- Search Engine Optimization (SEO)
- Code readability
- Website structure

Example

```
<nav>
```

```
<a href="index.html">Home</a>
```

```
<a href="about.html">About</a>
```

```
<a href="contact.html">Contact</a>
```

</nav>

Why Other Options Are Incorrect

Option	Reason
<div>	Generic container without semantic meaning.
<section>	Used for grouping related content, not navigation.
<header>	Represents introductory content, not specifically navigation.

Question 4: Identify the correct XML syntax and explain the error in other options.

Correct Answer: B

<note>

<to>User</to>

<from>Admin</from>

</note>

Explanation

XML requires:

- Every opening tag must have a matching closing tag.
- Tags must be properly nested.
- XML is case-sensitive.
- Documents must be well-formed.

Errors in Other Options

Option	Error
A	Missing closing </note> tag.
C	Tags are mismatched (<to> closed with </from>).
D	Missing closing angle bracket (>), making it invalid XML.

Correct XML Example

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<note>

  <to>User</to>

  <from>Admin</from>

</note>
```

Question 5: Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

Explanation

HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers (clients) and web servers.

It enables:

- Loading web pages
- Sending form data
- Requesting images and videos
- Downloading website resources

For secure communication, websites use HTTPS, which is HTTP with SSL/TLS encryption.

Comparison of Protocols

Protocol	Purpose
FTP	File transfer between systems
HTTP	Web page communication
SMTP	Sending email
SNMP	Network monitoring and management

Summary Table

Question	Correct Answer	Reason
Video Embedding	<video>	Standard HTML5 multimedia element
Form Submission	POST	Securely sends user data in request body
Navigation	<nav>	Semantic element for navigation links

Question	Correct Answer	Reason
XML Syntax	Option B	Properly closed and nested XML tags
Web Protocol	HTTP	Standard protocol for browser-server communication

Conclusion

The startup should adopt standard web technologies to ensure compatibility, security, and maintainability. HTML5 semantic elements like `<video>` and `<nav>` improve accessibility and page structure. The POST method securely submits user information, while properly formed XML ensures reliable configuration processing. Finally, HTTP/HTTPS enables efficient communication between clients and servers, forming the foundation of modern web applications.